15-Model COVID-19 Forecast Benchmark

Comparing Statistical, Machine Learning, Deep Learning, Foundation Models, and Conformal Recalibration

Time Series Analysis

2026-07-30

1 Introduction & Model Taxonomy

This document evaluates 15 time series forecasting models for weekly reported U.S. COVID-19 deaths between June 2020 and March 2023. We conduct rolling origin cross-validation across 140 forecast origin dates, measuring accuracy and calibration over 1- to 4-week horizons ($h \in \{1, 2, 3, 4\}$). To assess prediction interval quality, we evaluate coverage and interval scoring across four central nominal levels: 50%, 80%, 90%, and 95% confidence intervals ($\alpha \in \{0.50, 0.20, 0.10, 0.05\}$).

1.1 Evaluated Models

The benchmark encompasses five primary modeling paradigms summarized in the table below:

Model	Paradigm	Description
CDC CovidHub Ensemble	CDC Benchmark	Prospective multi-model ensemble benchmark submitted to the CDC COVID-19 Forecast Hub.
Empirical Baseline	Baseline	Persistence point forecast ($y_{t+h} = y_t$) with prediction intervals constructed from historical differences.
ARIMA(1,1,0)	Statistical	First-differenced autoregressive model with 1 lag and a linear trend.

Model	Paradigm	Description
ARIMA(2,1,0)	Statistical	First-differenced autoregressive model with 2 lags and a linear trend.
Holt ETS	Statistical	Exponential smoothing state-space model with additive trend.
Damped ETS	Statistical	Exponential smoothing state-space model with damped additive trend.
ARIMAX (ARIMA+Cases)	Exogenous	Autoregressive model on differenced log-deaths using 4-week lagged reported COVID-19 cases.
LightGBM (skforecast)	Machine Learning	Gradient boosted decision trees fit recursively over rolling lag features using `skforecast`.
XGBoost (sktime)	Machine Learning	Extreme gradient boosting fit recursively over autoregressive lag features using `sktime`.
N-BEATS (Neural)	Deep Learning	Deep neural architecture based on backward and forward residual basis expansion.
N-HiTS (Neural)	Deep Learning	Neural architecture using multi-rate sampling and hierarchical interpolation.
TimesFM (Google FM)	Foundation Model	Google's pretrained zero-shot foundation model for time series forecasting.
Chronos (Amazon FM)	Foundation Model	Amazon's pretrained zero-shot language-model-based time series forecaster.
ARIMAX + MAPIE (Conformal)	Recalibration	Conformal residual recalibration applied on top of ARIMAX point forecasts.

Model	Paradigm	Description
ARIMAX + MultiQT (Tracking)	Recalibration	Online multi-quantile tracking recalibration applied on top of ARIMAX point forecasts.

```
# Load dataset
cases_deaths = pd.read_csv("cases_deaths.csv")
covidhub_fc = pd.read_csv("covidhub_fc.csv")

cases_deaths["date"] = pd.to_datetime(cases_deaths["date"])
covidhub_fc["target_date"] = pd.to_datetime(covidhub_fc["target_date"])
covidhub_fc["forecast_date"] = covidhub_fc["target_date"] - pd.to_timedelta(covidhub_fc["h"]

cases_deaths_sorted = cases_deaths.sort_values("date").reset_index(drop=True)
```

2 Cross-Validation Dataset Setup

```
# Load 15-model cross-validation dataset via helper function
df_all = run_multi_alpha_cross_validation_pipeline(cases_deaths_sorted, covidhub_fc)

models = df_all["model"].unique().tolist()

colors = {
    "CDC CovidHub Ensemble": "#FFB800",
    "Empirical Baseline": "#7F8C8D",
    "ARIMA(1,1,0)": "#E74C3C",
    "ARIMA(2,1,0)": "#2980B9",
    "Holt ETS": "#2ECC71",
    "Damped ETS": "#8E44AD",
    "ARIMAX (ARIMA+Cases)": "#E91E63",
    "LightGBM (skforecast)": "#16A085",
    "XGBoost (sktime)": "#D35400",
    "N-BEATS (Neural)": "#9B59B6",
    "N-HiTS (Neural)": "#8E44AD",
    "TimesFM (Google FM)": "#34495E",
    "Chronos (Amazon FM)": "#F39C12",
    "ARIMAX + MAPIE (Conformal)": "#3498DB",
    "ARIMAX + MultiQT (Tracking)": "#9B1B8C"
```

```
}

# Compute detailed metrics dataframe across all horizons and alphas
metrics_df = compute_metrics_by_horizon(df_all, models, horizons=[1, 2, 3, 4])
```

3 Benchmark Metric Results

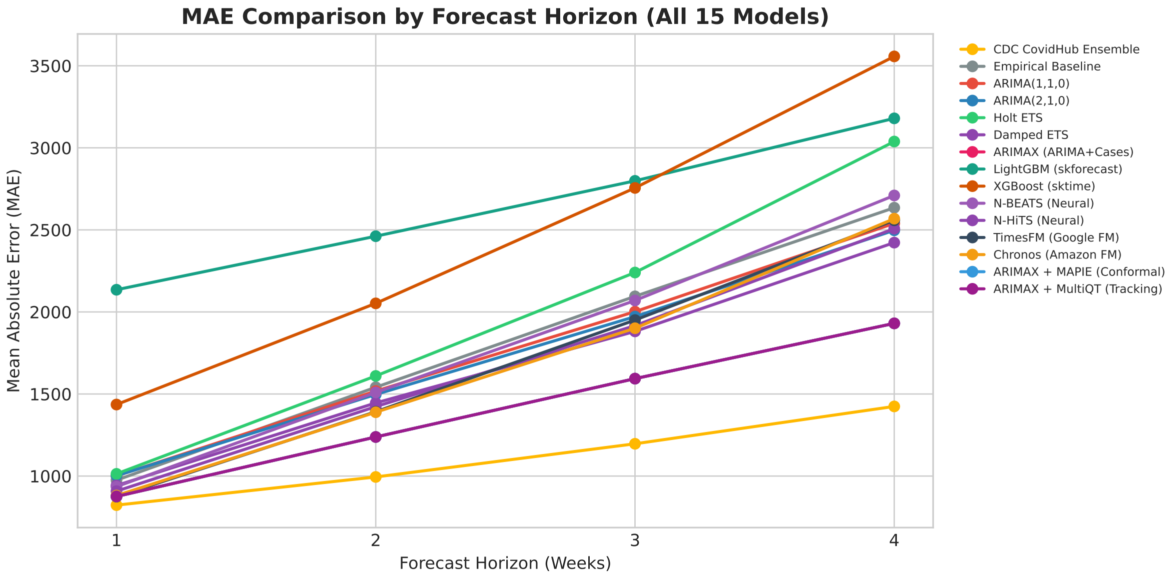
3.1 Point Forecast Accuracy (MAE)

We evaluate point forecast performance using Mean Absolute Error (MAE) across 1- to 4-week horizons. The table below shows MAE values for each model alongside their overall mean across all horizons.

```
mae_pivot = metrics_df.pivot(index="Model", columns="Horizon (h)", values="MAE")
mae_pivot["Overall Mean MAE"] = mae_pivot.mean(axis=1)
mae_pivot = mae_pivot.sort_values("Overall Mean MAE")
mae_pivot.round(2)
```

Horizon (h) Model	1	2	3	4	Overall Mean MAE
CDC CovidHub Ensemble	822.78	994.51	1196.60	1424.33	1109.56
ARIMAX (ARIMA+Cases)	875.70	1238.12	1593.64	1930.36	1409.45
ARIMAX + MAPIE (Conformal)	875.70	1238.12	1593.64	1930.36	1409.45
ARIMAX + MultiQT (Tracking)	875.70	1238.12	1593.64	1930.36	1409.45
Damped ETS	942.43	1445.63	1881.57	2422.39	1673.00
Chronos (Amazon FM)	881.33	1388.85	1900.75	2568.39	1684.83
N-HiTS (Neural)	908.01	1422.99	1916.69	2507.33	1688.76
TimesFM (Google FM)	874.43	1391.97	1950.56	2558.47	1693.86
ARIMA(2,1,0)	1001.90	1497.09	1971.54	2497.38	1741.98
ARIMA(1,1,0)	1000.94	1517.90	2001.79	2539.59	1765.06
N-BEATS (Neural)	935.39	1509.56	2069.73	2709.74	1806.10
Empirical Baseline	974.89	1541.07	2095.01	2635.04	1811.50
Holt ETS	1013.40	1610.03	2240.16	3038.36	1975.49
XGBoost (sktime)	1435.73	2052.27	2755.61	3557.12	2450.18
LightGBM (skforecast)	2135.23	2461.40	2798.81	3179.58	2643.76

```
# Plot MAE Curves using helper function
plot_metric_by_horizon(
    metrics_df,
    metric_col="MAE",
    title="MAE Comparison by Forecast Horizon (All 15 Models)",
    ylabel="Mean Absolute Error (MAE)",
    colors=colors
)
```


MAE Comparison by Forecast Horizon (All 15 Models)
Mean Absolute Error (MAE)
Forecast Horizon (Weeks)
CDC CovidHub Ensemble
Empirical Baseline
ARIMA(1,1,0)
ARIMA(2,1,0)
Holt ETS
Damped ETS
ARIMAX (ARIMA+Cases)
LightGBM (skforecast)
XGBoost (sktime)
N-BEATS (Neural)
N-HiTS (Neural)
TimesFM (Google FM)
Chronos (Amazon FM)
ARIMAX + MAPIE (Conformal)
ARIMAX + MultiQT (Tracking)

3.2 Prediction Interval Coverage and Calibration

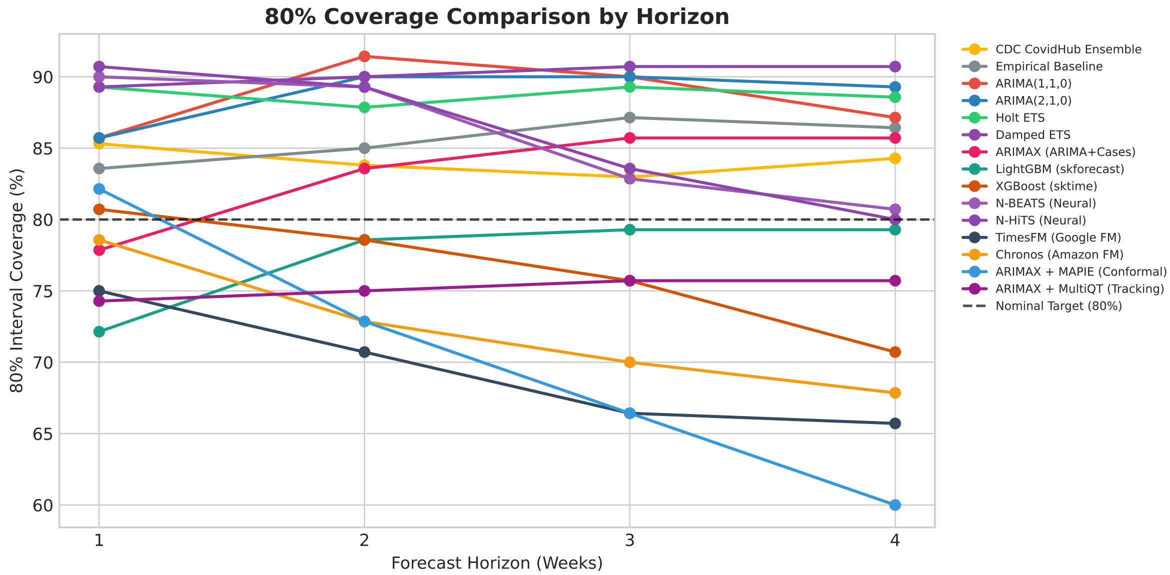
3.2.1 80% Nominal Coverage

To measure calibration, we examine whether empirical coverage aligns with nominal targets. The table below lists 80% prediction interval coverage across forecast horizons.

```
cov80_pivot = metrics_df.pivot(index="Model", columns="Horizon (h)", values="Coverage_80")
cov80_pivot["Overall 80% Coverage"] = cov80_pivot.mean(axis=1)
cov80_pivot = cov80_pivot.loc[models]
(cov80_pivot * 100).round(1).astype(str) + "%"
```

Horizon (h) Model	1	2	3	4	Overall 80% Coverage
CDC CovidHub Ensemble	85.3%	83.8%	83.0%	84.3%	84.1%
Empirical Baseline	83.6%	85.0%	87.1%	86.4%	85.5%
ARIMA(1,1,0)	85.7%	91.4%	90.0%	87.1%	88.6%
ARIMA(2,1,0)	85.7%	90.0%	90.0%	89.3%	88.8%
Holt ETS	89.3%	87.9%	89.3%	88.6%	88.8%
Damped ETS	89.3%	90.0%	90.7%	90.7%	90.2%
ARIMAX (ARIMA+Cases)	77.9%	83.6%	85.7%	85.7%	83.2%
LightGBM (skforecast)	72.1%	78.6%	79.3%	79.3%	77.3%
XGBoost (sktime)	80.7%	78.6%	75.7%	70.7%	76.4%
N-BEATS (Neural)	90.0%	89.3%	82.9%	80.7%	85.7%
N-HiTS (Neural)	90.7%	89.3%	83.6%	80.0%	85.9%
TimesFM (Google FM)	75.0%	70.7%	66.4%	65.7%	69.5%
Chronos (Amazon FM)	78.6%	72.9%	70.0%	67.9%	72.3%
ARIMAX + MAPIE (Conformal)	82.1%	72.9%	66.4%	60.0%	70.4%
ARIMAX + MultiQT (Tracking)	74.3%	75.0%	75.7%	75.7%	75.2%

```
# Plot 80% Coverage Curves
plot_metric_by_horizon(
    metrics_df,
    metric_col="Coverage_80",
    title="80% Coverage Comparison by Horizon",
    ylabel="80% Interval Coverage (%)",
    colors=colors,
    target_val=0.80,
    is_percentage=True
)
```



3.2.2 Multi-Alpha Calibration (50%, 80%, 90%, 95%)

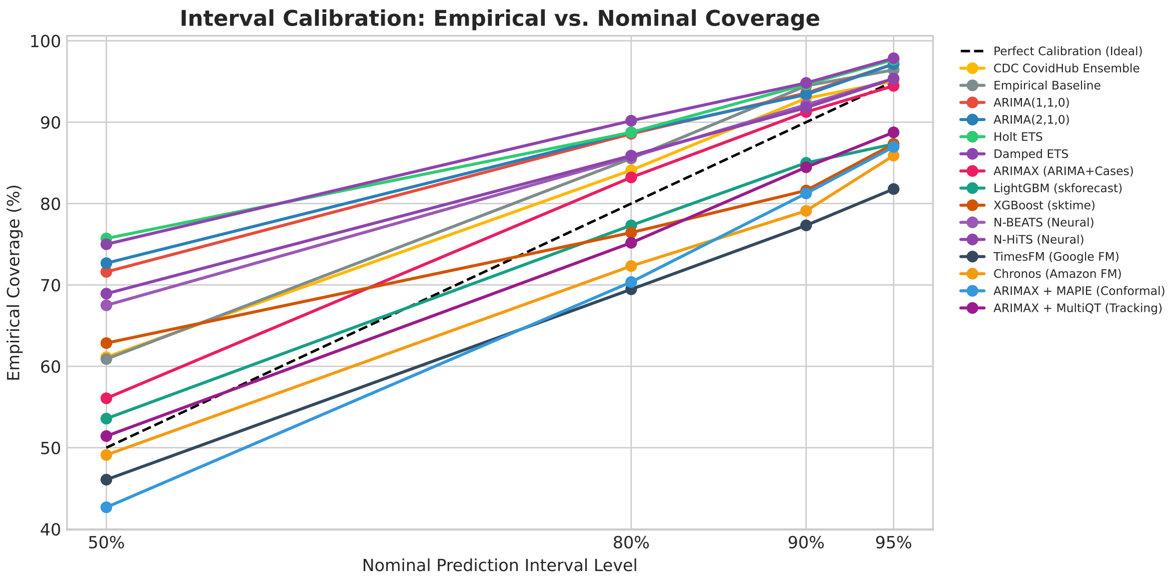
We summarize overall empirical coverage across all four nominal alpha levels to assess how well each model calibrates prediction bounds across the full central distribution.

```
multi_cov_summary = pd.DataFrame({
    "Model": models,
    "50% CI (Target 50%)": [metrics_df[metrics_df["Model"]==m]["Coverage_50"].mean() * 100 fo
    "80% CI (Target 80%)": [metrics_df[metrics_df["Model"]==m]["Coverage_80"].mean() * 100 fo
    "90% CI (Target 90%)": [metrics_df[metrics_df["Model"]==m]["Coverage_90"].mean() * 100 fo
    "95% CI (Target 95%)": [metrics_df[metrics_df["Model"]==m]["Coverage_95"].mean() * 100 fo
}).set_index("Model")
multi_cov_summary.round(1).astype(str) + "%"
```

Model	50% CI (Target 50%)	80% CI (Target 80%)	90% CI (Target 90%)	95
CDC CovidHub Ensemble	61.1%	84.1%	92.9%	95.
Empirical Baseline	60.9%	85.5%	94.5%	96.
ARIMA(1,1,0)	71.6%	88.6%	93.6%	97.
ARIMA(2,1,0)	72.7%	88.8%	93.4%	97.
Holt ETS	75.7%	88.8%	94.6%	97.
Damped ETS	75.0%	90.2%	94.8%	97.
ARIMAX (ARIMA+Cases)	56.1%	83.2%	91.3%	94.

Model	50% CI (Target 50%)	80% CI (Target 80%)	90% CI (Target 90%)	95
LightGBM (skforecast)	53.6%	77.3%	85.0%	87.
XGBoost (sktime)	62.9%	76.4%	81.6%	87.
N-BEATS (Neural)	67.5%	85.7%	92.1%	95.
N-HiTS (Neural)	68.9%	85.9%	91.8%	95.
TimesFM (Google FM)	46.1%	69.5%	77.3%	81.
Chronos (Amazon FM)	49.1%	72.3%	79.1%	85.
ARIMAX + MAPIE (Conformal)	42.7%	70.4%	81.2%	87.
ARIMAX + MultiQT (Tracking)	51.4%	75.2%	84.5%	88.

```
# Plot Calibration Across Alphas
plot_coverage_by_alpha(metrics_df, models, colors)
```



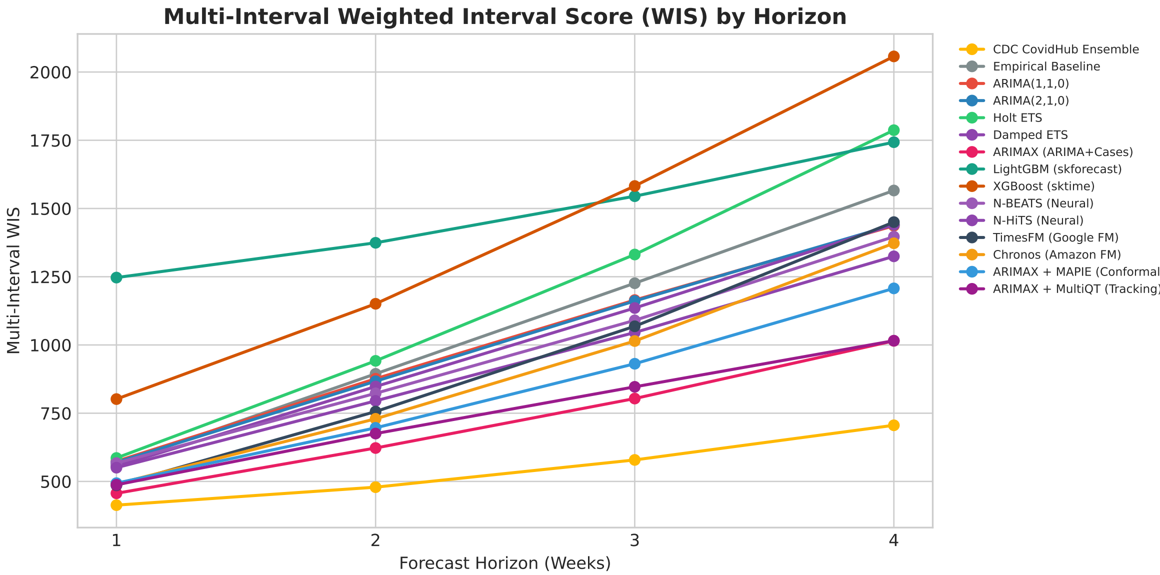
3.3 Probabilistic Interval Scoring (WIS)

The Weighted Interval Score (WIS) combines point error with penalties for interval width and out-of-bounds observations. Multi-interval WIS averages these scores across all four central prediction intervals.

```
wis_multi_pivot = metrics_df.pivot(index="Model", columns="Horizon (h)", values="WIS_Multi")
wis_multi_pivot["Overall Multi-WIS"] = wis_multi_pivot.mean(axis=1)
wis_multi_pivot = wis_multi_pivot.sort_values("Overall Multi-WIS")
wis_multi_pivot.round(2)
```

Horizon (h) Model	1	2	3	4	Overall Multi-WIS
CDC CovidHub Ensemble	413.17	478.98	578.79	705.73	544.17
ARIMAX (ARIMA+Cases)	456.47	622.43	803.67	1015.09	724.41
ARIMAX + MultiQT (Tracking)	487.73	675.36	846.62	1015.67	756.35
ARIMAX + MAPIE (Conformal)	493.64	696.32	931.15	1207.32	832.11
Chronos (Amazon FM)	491.22	729.44	1014.41	1372.62	901.92
N-HiTS (Neural)	550.40	795.16	1045.69	1324.75	929.00
TimesFM (Google FM)	485.79	756.10	1068.48	1450.11	940.12
N-BEATS (Neural)	567.29	822.83	1090.18	1396.71	969.25
Damped ETS	555.65	847.67	1135.37	1440.98	994.91
ARIMA(2,1,0)	570.26	867.03	1161.13	1440.37	1009.70
ARIMA(1,1,0)	573.94	876.24	1164.52	1436.71	1012.85
Empirical Baseline	563.39	894.26	1226.19	1565.94	1062.44
Holt ETS	585.91	941.77	1331.38	1786.90	1161.49
XGBoost (sktime)	801.71	1150.86	1582.38	2057.25	1398.05
LightGBM (skforecast)	1246.90	1374.34	1545.04	1742.89	1477.29

```
# Plot Multi-Interval WIS Curves
plot_metric_by_horizon(
    metrics_df,
    metric_col="WIS_Multi",
    title="Multi-Interval Weighted Interval Score (WIS) by Horizon",
    ylabel="Multi-Interval WIS",
    colors=colors
)
```



Multi-Interval Weighted Interval Score (WIS) by Horizon
Multi-Interval WIS
Forecast Horizon (Weeks)
500
750
1000
1250
1500
1750
2000
1
2
3
4
CDC CovidHub Ensemble
Empirical Baseline
ARIMA(1,1,0)
ARIMA(2,1,0)
Holt ETS
Damped ETS
ARIMAX (ARIMA+Cases)
LightGBM (skforecast)
XGBoost (sktime)
N-BEATS (Neural)
N-HiTS (Neural)
TimesFM (Google FM)
Chronos (Amazon FM)
ARIMAX + MAPIE (Conformal)
ARIMAX + MultiQT (Tracking)

3.4 Comprehensive Model Leaderboard

The leaderboard ranks all 15 models by overall Multi-Interval WIS, providing a combined view of point accuracy, interval sharpness, and coverage calibration.

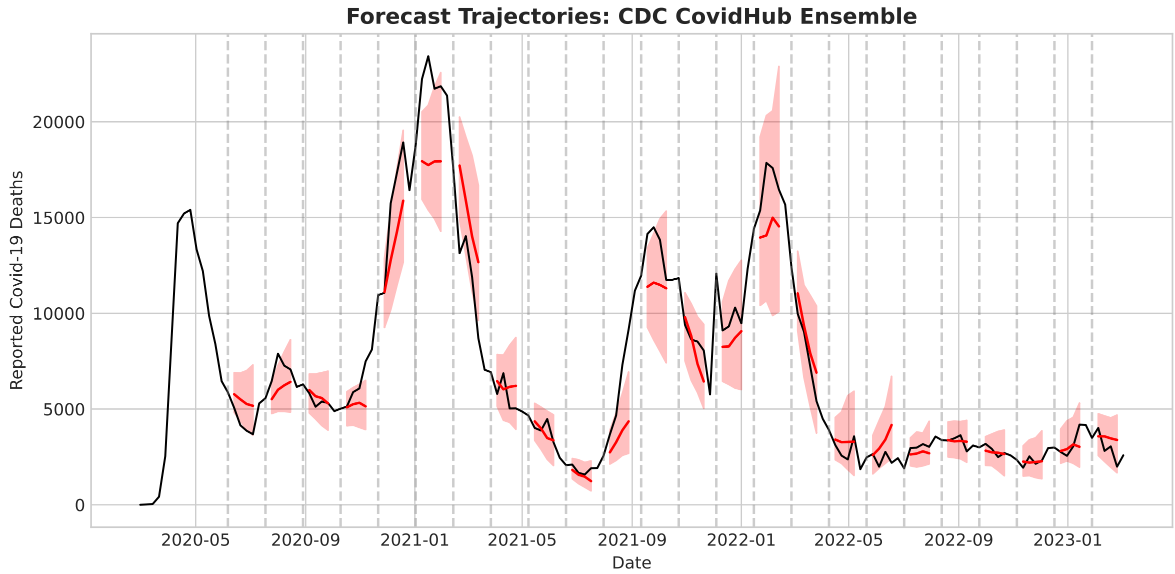
```
summary_leaderboard = pd.DataFrame({
    "Model": models,
    "Overall MAE": [metrics_df[metrics_df["Model"]==m]["MAE"].mean() for m in models],
    "Multi-WIS": [metrics_df[metrics_df["Model"]==m]["WIS_Multi"].mean() for m in models],
    "80% WIS": [metrics_df[metrics_df["Model"]==m]["WIS_80"].mean() for m in models],
    "50% Coverage": [metrics_df[metrics_df["Model"]==m]["Coverage_50"].mean() for m in models
    "80% Coverage": [metrics_df[metrics_df["Model"]==m]["Coverage_80"].mean() for m in models
    "90% Coverage": [metrics_df[metrics_df["Model"]==m]["Coverage_90"].mean() for m in models
    "95% Coverage": [metrics_df[metrics_df["Model"]==m]["Coverage_95"].mean() for m in models
}).set_index("Model").sort_values("Multi-WIS")

summary_leaderboard.round(2)
```

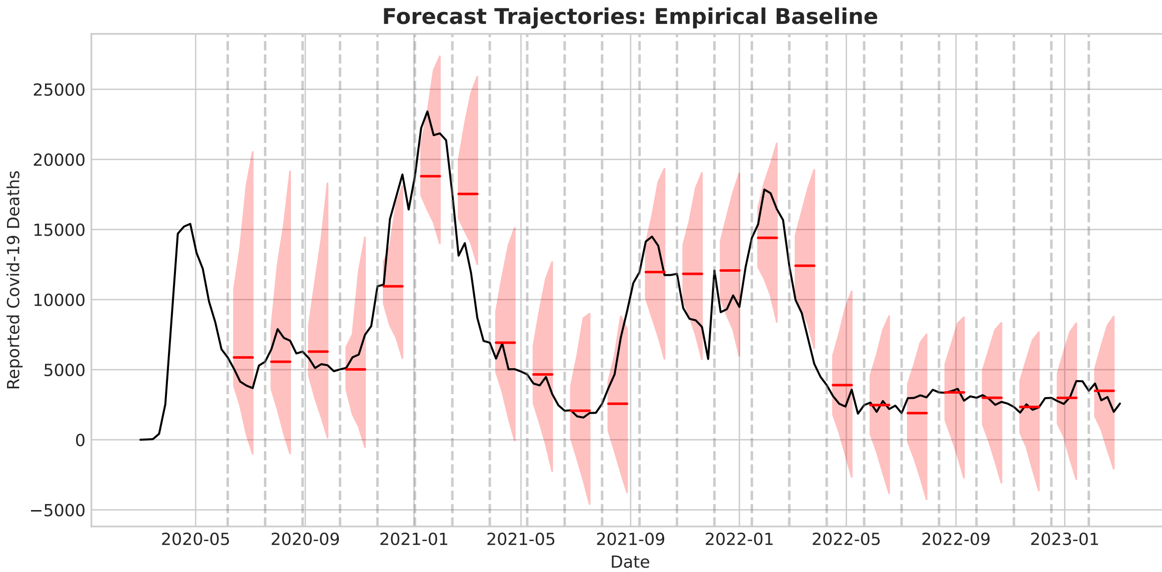
Model	Overall MAE	Multi-WIS	80% WIS	50% Coverage	80% Coverage	9
CDC CovidHub Ensemble	1109.56	544.17	1063.21	0.61	0.84	0
ARIMAX (ARIMA+Cases)	1409.45	724.41	1391.46	0.56	0.83	0
ARIMAX + MultiQT (Tracking)	1409.45	756.35	1448.25	0.51	0.75	0
ARIMAX + MAPIE (Conformal)	1409.45	832.11	1557.26	0.43	0.70	0
Chronos (Amazon FM)	1684.83	901.92	1714.01	0.49	0.72	0
N-HiTS (Neural)	1688.76	929.00	1770.69	0.69	0.86	0
TimesFM (Google FM)	1693.86	940.12	1762.06	0.46	0.69	0
N-BEATS (Neural)	1806.10	969.25	1853.76	0.68	0.86	0
Damped ETS	1673.00	994.91	1859.09	0.75	0.90	0
ARIMA(2,1,0)	1741.98	1009.70	1900.20	0.73	0.89	0
ARIMA(1,1,0)	1765.06	1012.85	1909.19	0.72	0.89	0
Empirical Baseline	1811.50	1062.44	2007.65	0.61	0.86	0
Holt ETS	1975.49	1161.49	2188.00	0.76	0.89	0
XGBoost (sktime)	2450.18	1398.05	2618.95	0.63	0.76	0
LightGBM (skforecast)	2643.76	1477.29	2757.75	0.54	0.77	0

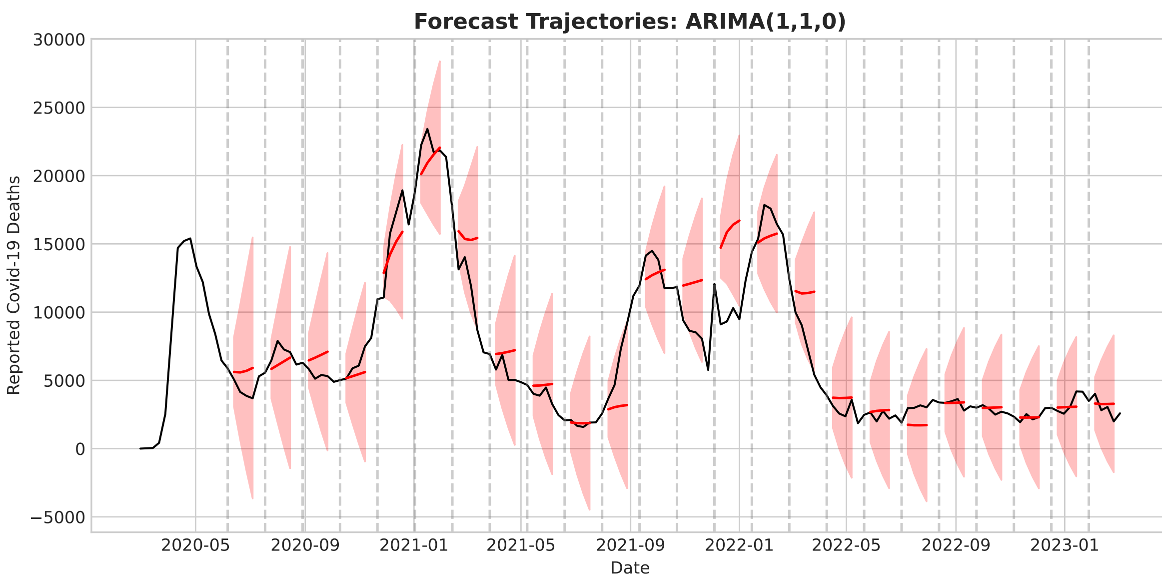
4 Forecast Trajectories

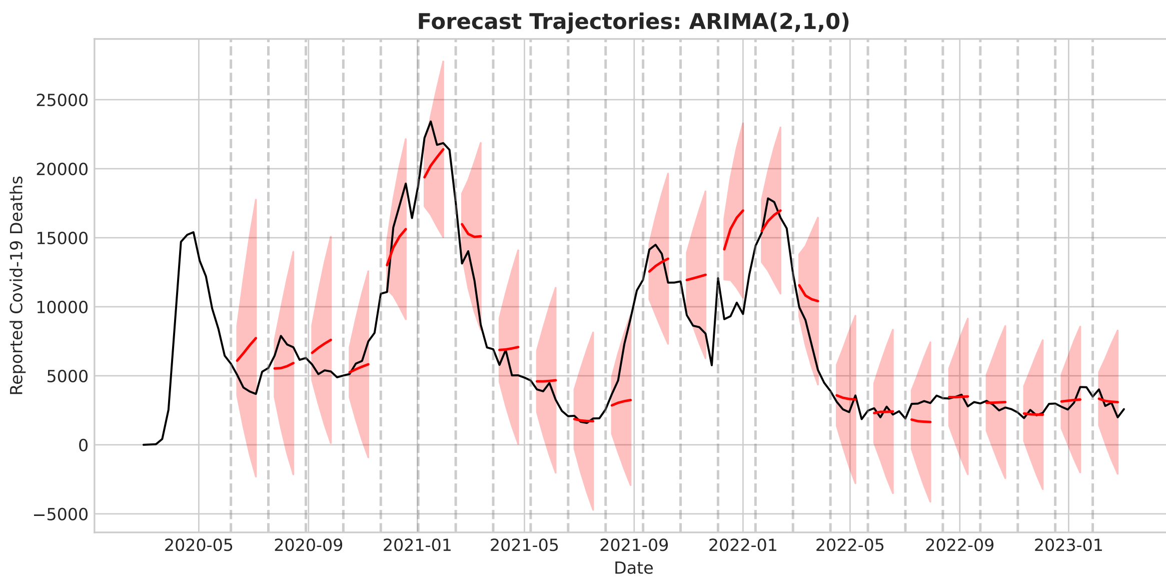
We visualize the sequential forecast trajectories produced by each model at six-week intervals across the evaluation period.

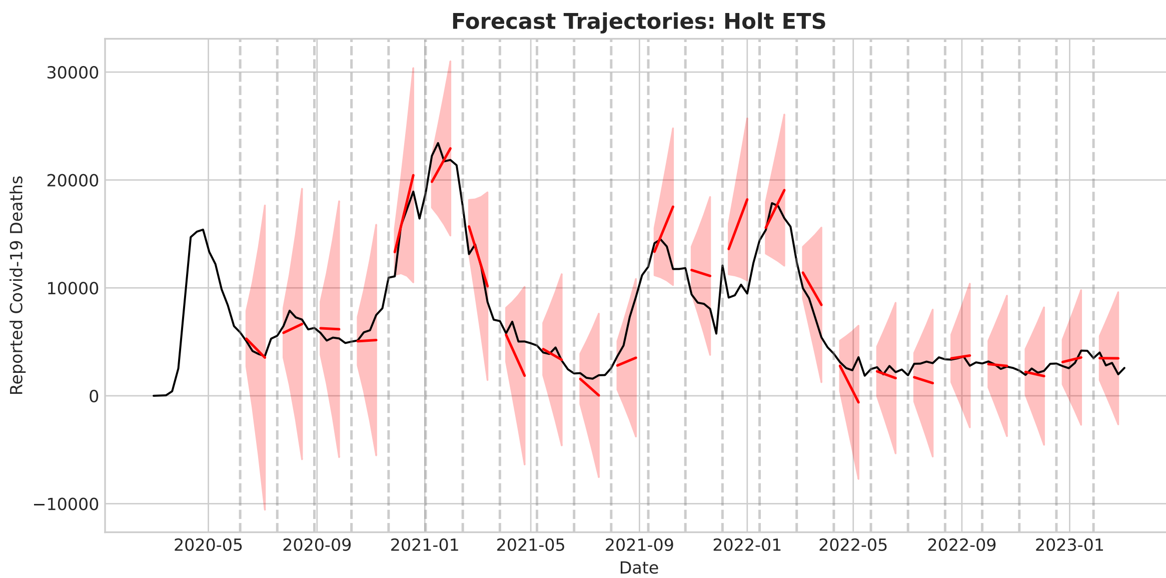
```
fc_dates_all = cases_deaths_sorted[(cases_deaths_sorted["date"] >= pd.to_datetime("2020-06-0
fc_dates_plot = pd.to_datetime(fc_dates_all)[::6]

for model in models:
    plot_forecast_trajectories(
        cases_deaths_sorted,
        df_all[df_all["model"] == model],
        f"Forecast Trajectories: {model}",
        fc_dates=fc_dates_plot
    )
```

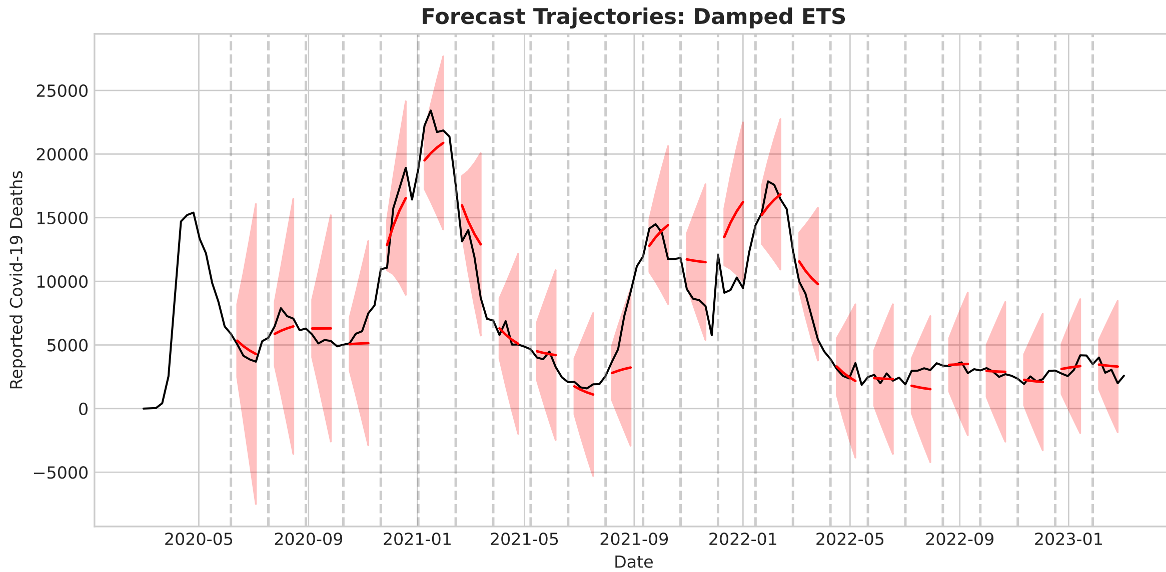


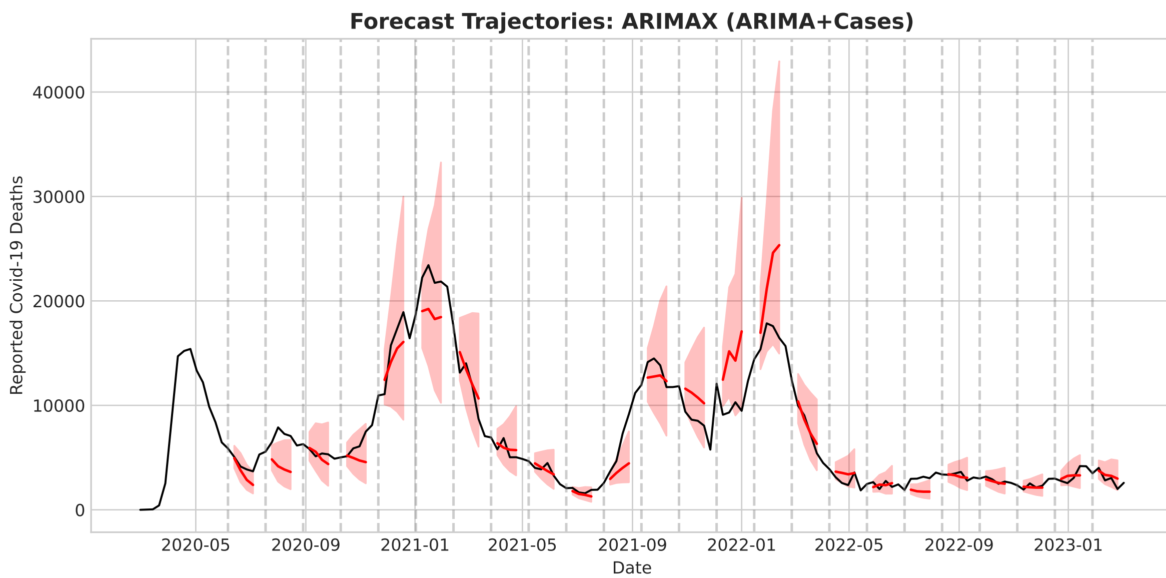

Forecast Trajectories: Empirical Baseline
Reported Covid-19 Deaths
Date
25000
20000
15000
10000
5000
0
−5000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

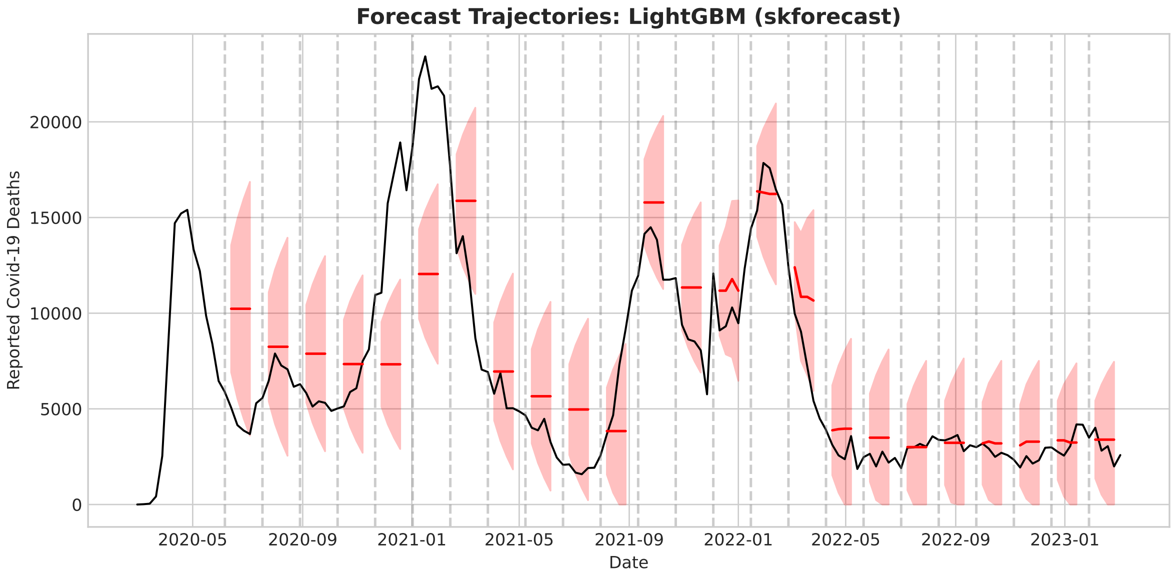

Forecast Trajectories: ARIMA(1,1,0)
Reported Covid-19 Deaths
Date
30000
25000
20000
15000
10000
5000
0
−5000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01


Forecast Trajectories: ARIMA(2,1,0)
Reported Covid-19 Deaths
25000
20000
15000
10000
5000
0
−5000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date

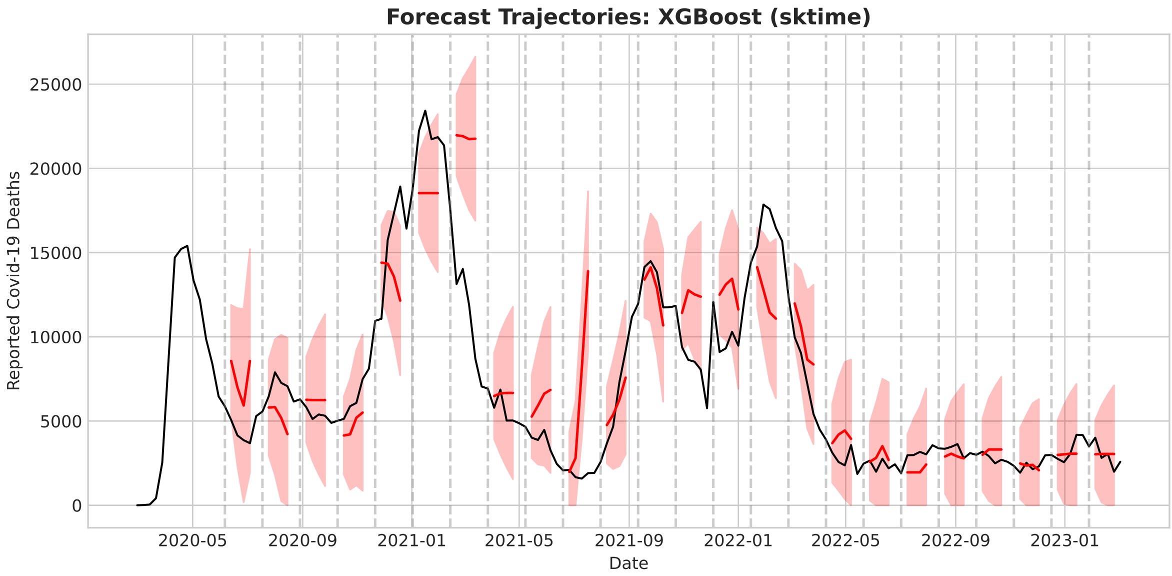

Forecast Trajectories: Holt ETS
Reported Covid-19 Deaths
30000
20000
10000
0
−10000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date


Forecast Trajectories: Damped ETS
Reported Covid-19 Deaths
Date
25000
20000
15000
10000
5000
0
−5000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

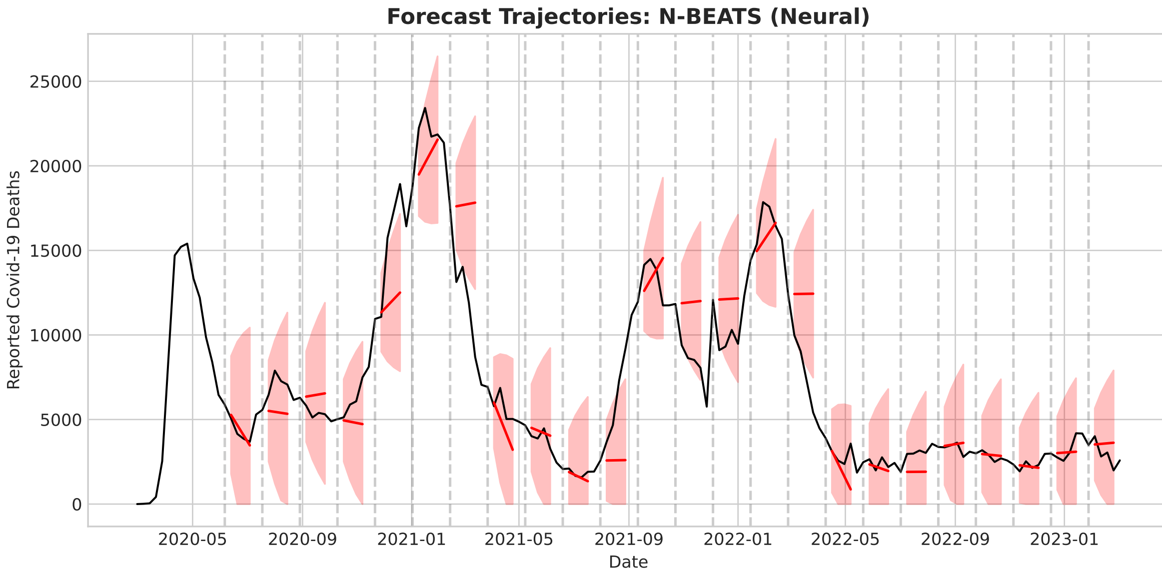

Forecast Trajectories: ARIMAX (ARIMA+Cases)
Reported Covid-19 Deaths
Date
40000
30000
20000
10000
0
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

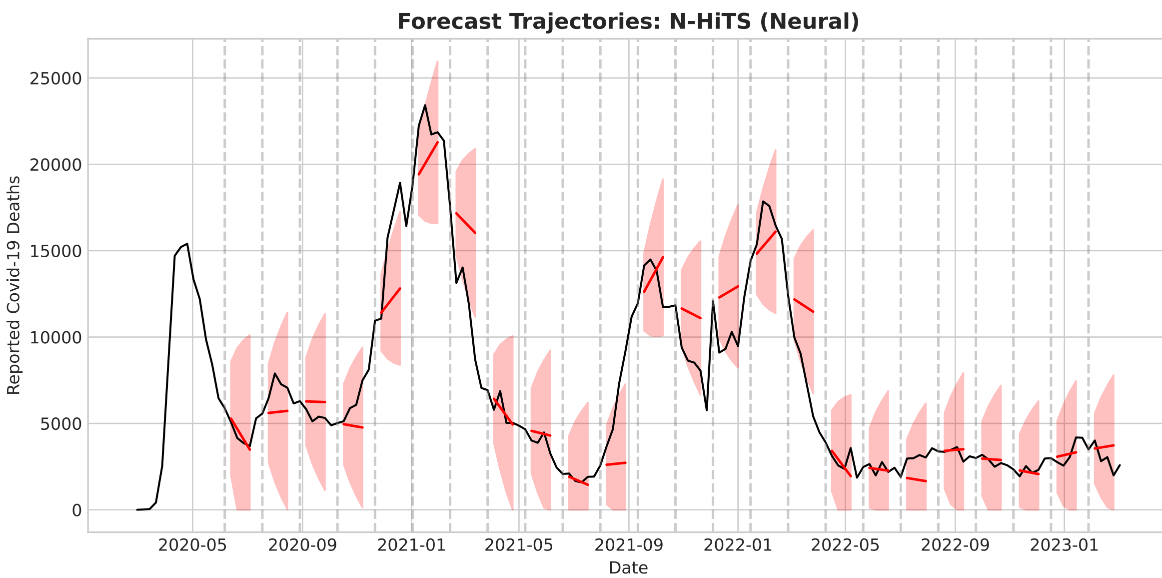


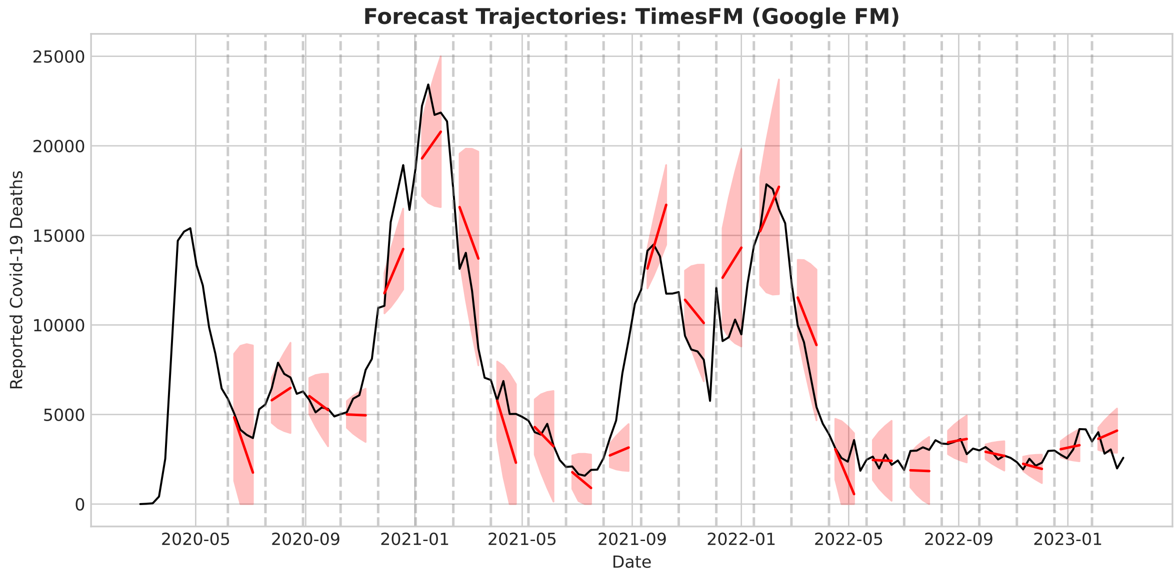
Forecast Trajectories: LightGBM (skforecast)
Reported Covid-19 Deaths
Date
0
5000
10000
15000
20000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

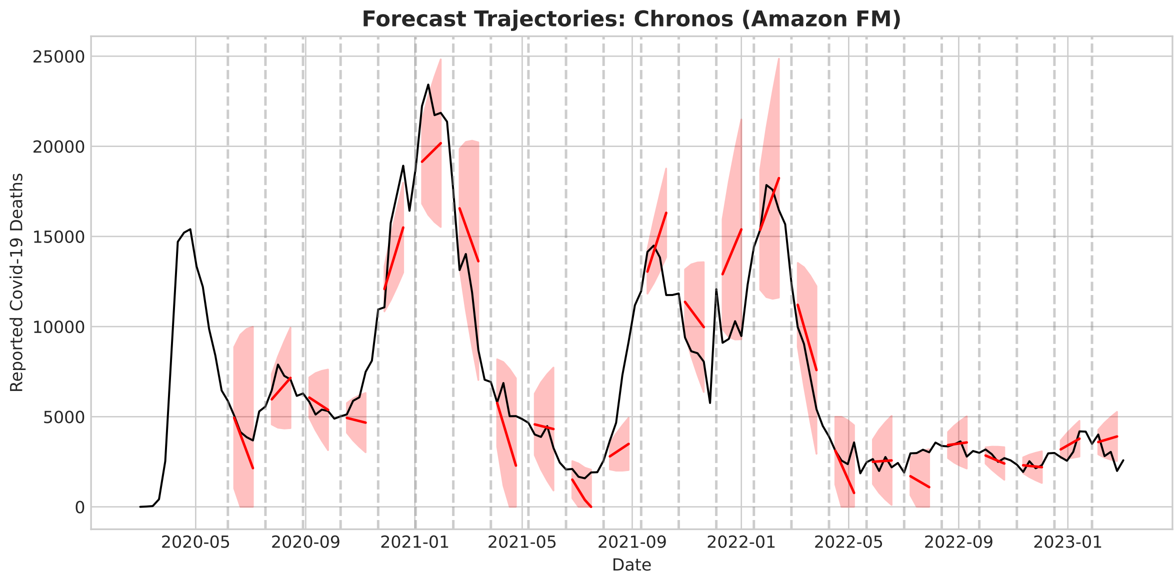


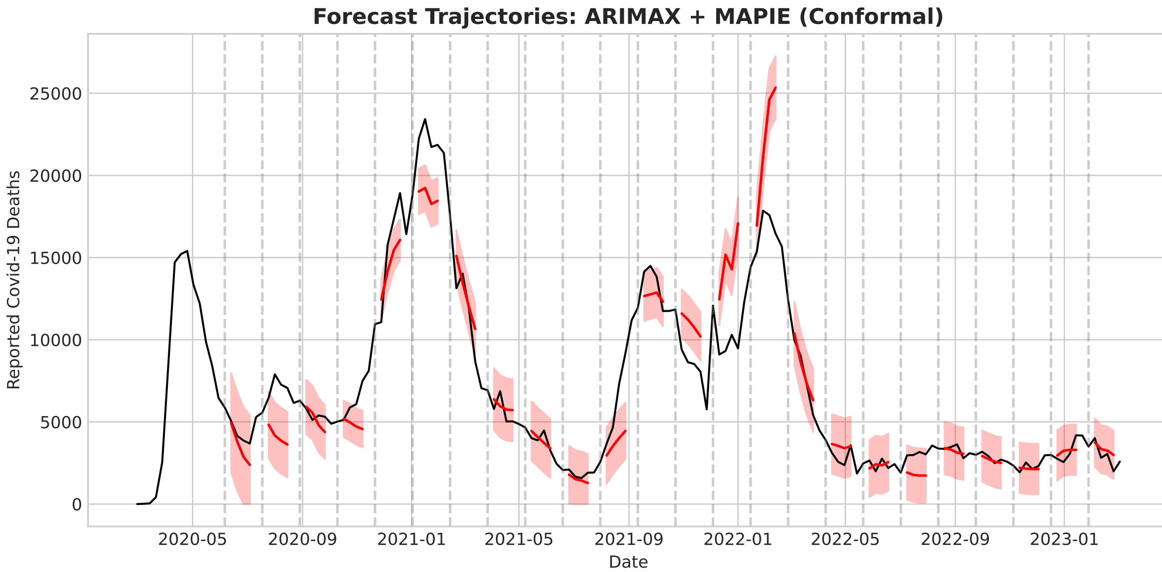
Forecast Trajectories: XGBoost (sktime)
Reported Covid-19 Deaths
Date
0
5000
10000
15000
20000
25000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01

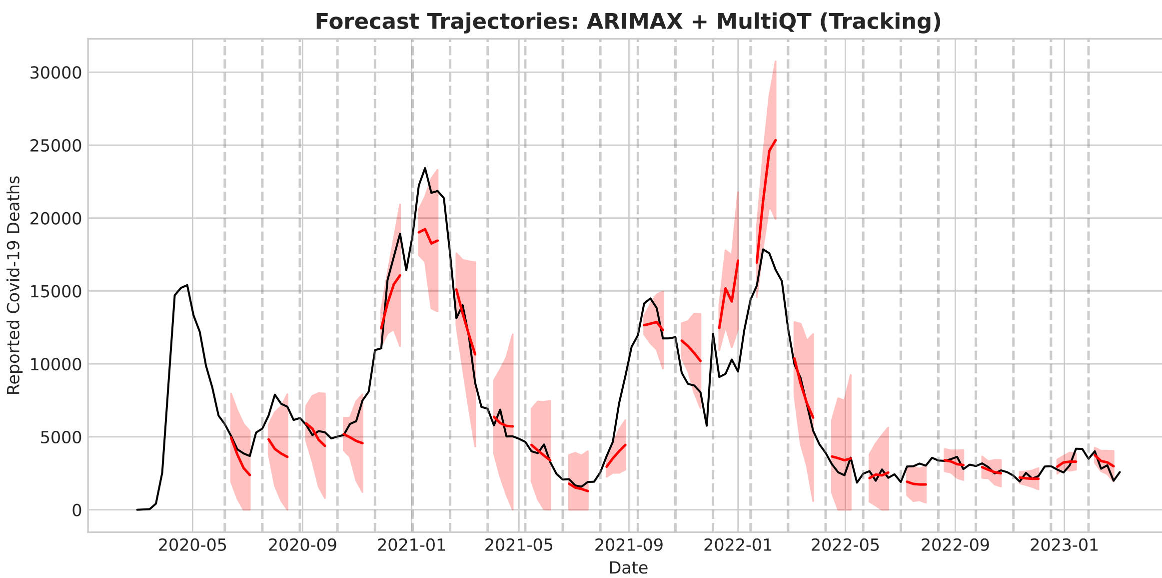

Forecast Trajectories: N-BEATS (Neural)
Reported Covid-19 Deaths
25000
20000
15000
10000
5000
0
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date


Forecast Trajectories: N-HiTS (Neural)
Reported Covid-19 Deaths
25000
20000
15000
10000
5000
0
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date


Forecast Trajectories: TimesFM (Google FM)
Reported Covid-19 Deaths
25000
20000
15000
10000
5000
0
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date


Forecast Trajectories: Chronos (Amazon FM)
Reported Covid-19 Deaths
25000
20000
15000
10000
5000
0
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date


Forecast Trajectories: ARIMAX + MAPIE (Conformal)
Reported Covid-19 Deaths
0
5000
10000
15000
20000
25000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date


Forecast Trajectories: ARIMAX + MultiQT (Tracking)
Reported Covid-19 Deaths
0
5000
10000
15000
20000
25000
30000
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Date